



Exploratory Data Analysis (EDA) Report – Geldium Delinquency Dataset

1. Introduction

This report presents an exploratory analysis of Geldium’s customer dataset to uncover behavioral and financial patterns linked to loan delinquency. The aim is to assess the dataset’s integrity, identify influential variables, and lay the groundwork for predictive modeling. By addressing data gaps and surfacing risk factors, this analysis supports the development of a reliable delinquency prediction framework.

2. Dataset Overview

The dataset comprises over 500 customer records, each containing demographic, financial, and behavioural attributes.

Highlights:

- **Total records:** 500+
- **Key variables:**
 - Credit_Utilization: Proportion of credit used
 - Missed_Payments: Count of payment defaults
 - Debt_to_Income_Ratio: Financial stress indicator
 - Credit_Score: Creditworthiness metric
 - Monthly Payment History (Month_1 to Month_6): Behavioral trend
 - Delinquent_Account: Target variable (binary)

Data types:

- **Numerical:** Age, Income, Loan_Balance, Credit_Score, Debt_to_Income_Ratio
- **Categorical:** Employment_Status, Credit_Card_Type, Location, Monthly Payment History

Notable issues:

- Credit_Utilization values exceeding 1.0 suggest data entry errors
- Employment labels vary widely (e.g., “EMP”, “Self-employed”, “employed”)
- Some records show zero loan balance despite missed payments

3. Missing Data Analysis

Identified gaps:

- Income: Missing in several entries
- Credit_Utilization: Contains invalid or missing values
- Monthly Payment History: Incomplete sequences across months

Treatment decisions:

Feature	Method	Reasoning
Credit_Utilization	Median Imputation	Reduces skew and maintains distribution integrity
Income	Synthetic Generation	Based on age, employment, and location for realistic values
Monthly Payment History	Mode Imputation	Preserves behavioral consistency across categorical entries

4. Key Findings and Risk Indicators

Observed relationships:

- Customers with Credit_Utilization > 0.6 tend to have more missed payments
- Debt_to_Income_Ratio > 0.4 is frequently linked to delinquent accounts
- Credit_Score < 500 is a strong predictor of financial instability

Surprising insights:

- High-income individuals occasionally show erratic payment behavior
- Some young customers with high credit scores still default, suggesting thin credit files
- “Student” cardholders show higher delinquency rates than premium cardholders

Top risk indicators:

Indicator	Why It Matters
High Credit Utilization	Indicates financial strain and overextension
Frequent Missed Payments	Direct evidence of repayment issues
Low Credit Score	Reflects poor historical financial behaviour
High Debt-to-Income Ratio	Suggests limited repayment capacity
Unstable Employment Status	Associated with income volatility and increased risk

5. GenAI Integration & Prompt Strategy

Generative AI tools were used to streamline data review, imputation planning, and pattern detection. These tools helped surface nuanced insights and automate parts of the analysis.

Prompts used:

- “Which financial features show the strongest divergence between delinquent and non-delinquent customers?”
- “Generate synthetic income values for missing entries using employment and age as anchors.”
- “Cluster customers by payment behavior and highlight high-risk groups.”
- “Detect anomalies in credit utilization and suggest corrective strategies.”
- “Rank variables by predictive potential using correlation with delinquency.”

6. Conclusion & Next Steps

The dataset offers a rich mix of behavioral and financial signals for delinquency modeling. Key indicators such as credit utilization, missed payments, and debt-to-income ratio show strong predictive relevance. However, inconsistencies in employment labels and unrealistic values must be resolved to ensure model reliability.

Next steps:

- Finalize data cleaning and standardization
- Engineer behavioral features (e.g., payment volatility index)
- Build and test predictive models
- Evaluate fairness and performance across customer segments